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Week 3

SESSION-5

Task 1: To Apply Single Quotes, Preserve Literal Content, Ignore Variable Expansion, Store Quoted Strings, Validate Parsing Results, Test Edge Cases.

CODE:

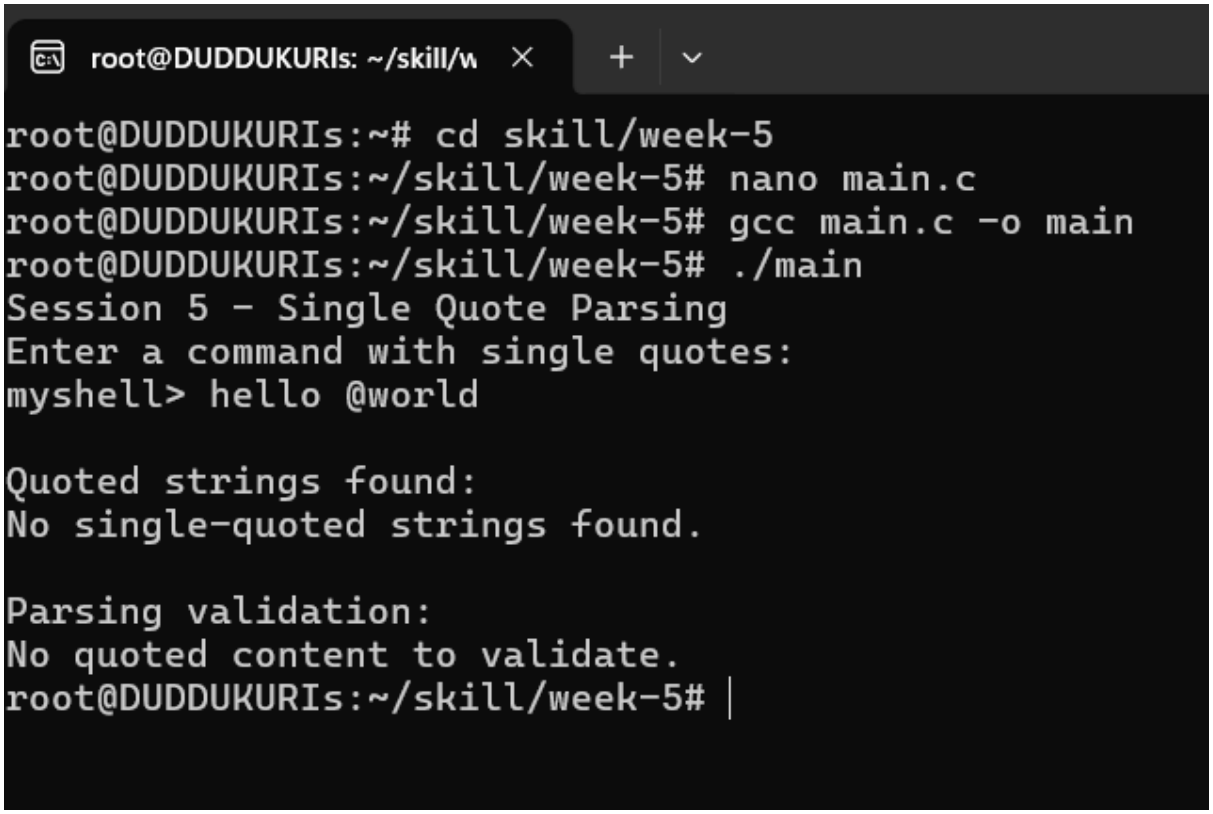
```
root@DUDDUKURIs: ~/skill/w
  GNU nano 7.2                                                main.c *
#include <stdio.h>
#include <string.h>

#define MAX_INPUT 200
#define MAX_QUOTES 20

int main() {
    char input[MAX_INPUT];
    char quoted_strings[MAX_QUOTES][MAX_INPUT];
    int quote_count = 0;

    printf("Session 5 - Single Quote Parsing\n");
    printf("Enter a command with single quotes:\n");
    printf("myshell> ");

    fgets(input, sizeof(input), stdin);

    /* Remove newline */
    input[strcspn(input, "\n")] = '\0';

    int i = 0;

    /* Parse input and find single-quoted strings */
    while (input[i] != '\0') {

        if (input[i] == '\'') {

            i++;

            int j = 0;
```

```
    /* Store everything until the closing quote */
    while (input[i] != '\0' && input[i] != '\'') {

        if (j < MAX_INPUT - 1) {
            quoted_strings[quote_count][j++] = input[i];
        }

        i++;
    }

    quoted_strings[quote_count][j] = '\0';
```

```
        /* Check for closing quote */
        if (input[i] == '\'') {
            if (quote_count < MAX_QUOTES - 1) {
                quote_count++;
            }
            i++;
        }
        else {
            printf("Error: Unclosed single quote.\n");
            return 1;
        }

    } else {
        i++;
    }
}
```

```
    printf("\nQuoted strings found:\n");

    if (quote_count == 0) {
        printf("No single-quoted strings found.\n");
    }
    else {
        for (int k = 0; k < quote_count; k++) {
            printf("Quoted String %d: [%s]\n",
                   k + 1, quoted_strings[k]);
        }
    }

    printf("\nParsing validation:\n");

    if (quote_count > 0) {
        printf("Single-quote parsing successful.\n");
    }
    else {
        printf("No quoted content to validate.\n");
    }

    return 0;
}
```

OUTPUT:



Task 2: To Apply Double Quotes, Preserve Spaces, Allow Variable Expansion, Parse Nested Tokens, Validate Outputs, Test Quoted Commands.

CODE:

root@DUDDUKURIs: ~/skill/w

GNU nano 7.2 main2.c *

```c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

#define MAX_INPUT 300
#define MAX_TOKENS 50
#define MAX_TOKEN_LENGTH 100

/* Expand environment variables such as $USER */
void expand_variable(char *input, char *output) {
    int i = 0;
    int j = 0;

    while (input[i] != '\0' && j < MAX_INPUT - 1) {

        if (input[i] == '$') {
            i++;

            char variable[50];
            int k = 0;

            /* Read variable name */
            while (isalnum((unsigned char)input[i]) ||
                   input[i] == '_') {

                if (k < 49) {
                    variable[k++] = input[i];
                }

                i++;
            }
```

```c
            variable[k] = '\0';

            /* Get environment variable value */
            char *value = getenv(variable);

            if (value != NULL) {
                int v = 0;

                while (value[v] != '\0' &&
                       j < MAX_INPUT - 1) {
```

```
                    output[j++] = value[v++];
                }
            }

        } else {
            output[j++] = input[i++];
        }
    }

    output[j] = '\0';
}

int main() {

    char input[MAX_INPUT];
    char expanded[MAX_INPUT];

    char tokens[MAX_TOKENS][MAX_TOKEN_LENGTH];

    int token_count = 0;
    int i = 0;
```

```
printf("Session 5 - Double Quote Parsing\n");
printf("Double quotes preserve spaces and allow variable expansion.\n");
printf("Enter a command:\n");
printf("myshell> ");

fgets(input, sizeof(input), stdin);

/* Remove newline */
input[strcspn(input, "\n")] = '\0';

/* Handle empty command */
if (strlen(input) == 0) {
    printf("Empty command.\n");
    return 0;
}

/* Parse input */
while (input[i] != '\0' &&
       token_count < MAX_TOKENS) {

    /* Skip spaces */
    while (isspace((unsigned char)input[i])) {
        i++;
    }

    if (input[i] == '\0') {
        break;
    }

    int j = 0;
```

```c
/* Double-quoted string */
if (input[i] == '"') {

    i++;

    while (input[i] != '\0' &&
           input[i] != '"') {

        if (j < MAX_TOKEN_LENGTH - 1) {
            tokens[token_count][j++] = input[i];
        }

        i++;
    }

    /* Check closing quote */
    if (input[i] == '"') {
        i++;
    } else {
        printf("Error: Unclosed double quote.\n");
        return 1;
    }

    tokens[token_count][j] = '\0';
    token_count++;

}

/* Normal token */
else {
```

```c
            while (input[i] != '\0' &&
                   !isspace((unsigned char)input[i]) &&
                   input[i] != '"') {

                if (j < MAX_TOKEN_LENGTH - 1) {
                    tokens[token_count][j++] = input[i];
                }

                i++;
            }

            tokens[token_count][j] = '\0';
```

```
            if (j > 0) {
                token_count++;
            }
        }
    }

    /* Display tokens before expansion */
    printf("\nTokens before variable expansion:\n");

    for (int k = 0; k < token_count; k++) {
        printf("Token %d: [%s]\n", k + 1, tokens[k]);
    }
```

```
    /* Variable expansion */
    printf("\nVariable expansion:\n");

    for (int k = 0; k < token_count; k++) {

        expand_variable(tokens[k], expanded);

        printf("Token %d: [%s]\n",
               k + 1, expanded);
    }

    /* Validation */
    printf("\nParsing validation:\n");
    printf("Double-quote parsing successful.\n");
    printf("Spaces inside quotes were preserved.\n");
    printf("Variable expansion was processed.\n");

    return 0;
}
```

OUTPUT:

```
root@DUDDUKURIs:~/skill/week-5# nano main2.c
root@DUDDUKURIs:~/skill/week-5# gcc main2.c -o main2
root@DUDDUKURIs:~/skill/week-5# ./main2
Session 5 - Double Quote Parsing
Double quotes preserve spaces and allow variable expansion.
Enter a command:
myshell> hello world

Tokens before variable expansion:
Token 1: [hello]
Token 2: [world]

Variable expansion:
Token 1: [hello]
Token 2: [world]

Parsing validation:
Double-quote parsing successful.
Spaces inside quotes were preserved.
Variable expansion was processed.
root@DUDDUKURIs:~/skill/week-5#
```

SESSION-6

Task 1: To Create Process Escape Sequences, Handle Escaped Spaces, Escape Special Symbols, Preserve Characters, Validate Parser Output, Test Complex Inputs.

CODE:

```
  GNU nano 7.2                                                  main.c *
#include <stdio.h>
#include <string.h>

#define MAX_INPUT 300

int main() {
    char input[MAX_INPUT];
    char output[MAX_INPUT];

    int i = 0;
    int j = 0;
    int escaped = 0;

    printf("Session 6 - Escape Sequence Parsing\n");
    printf("Enter a command with escaped spaces or special characters:\n");
    printf("myshell> ");

    fgets(input, sizeof(input), stdin);

    input[strcspn(input, "\n")] = '\0';

    /* Handle empty input */
    if (strlen(input) == 0) {
        printf("Empty input.\n");
        return 0;
    }
```

```
/* Process escape sequences */
while (input[i] != '\0' && j < MAX_INPUT - 1) {

    if (escaped) {
        output[j++] = input[i];
        escaped = 0;
    }
    else if (input[i] == '\\') {
        escaped = 1;
    }
    else {
        output[j++] = input[i];
    }

    i++;
}
```

```c
    /* Check if input ends with escape character */
    if (escaped) {
        printf("Warning: Escape character at the end of input.\n");
    }

    output[j] = '\0';

    printf("\nOriginal input:\n");
    printf("%s\n", input);

    printf("\nProcessed input:\n");
    printf("%s\n", output);
```

```c
    printf("\nParser validation:\n");
    printf("Escape sequences processed successfully.\n");
    printf("Escaped characters were preserved.\n");

    return 0;
}
```

OUTPUT:

```
root@DUDDUKURIs:~/skill/week-6# nano main.c
root@DUDDUKURIs:~/skill/week-6# gcc main.c -o main
root@DUDDUKURIs:~/skill/week-6# ./main
Session 6 - Escape Sequence Parsing
Enter a command with escaped spaces or special characters:
myshell> ls

Original input:
ls

Processed input:
ls

Parser validation:
Escape sequences processed successfully.
Escaped characters were preserved.
root@DUDDUKURIs:~/skill/week-6#
```

Task 2: To Create Child Processes, Execute Programs, Handle Execution Errors, Pass Arguments, Manage Parent Process, Test Command Launching

CODE:

```
 GNU nano 7.2                                                main2.c *
#include <stdio.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid;
    int status;

    printf("Session 6 - Process Execution\n");

    pid = fork();

    if (pid < 0) {
        perror("fork failed");
        return 1;
    }

    if (pid == 0) {
        /* Child process */
        printf("\nChild Process\n");
        printf("Child PID: %d\n", getpid());
        printf("Parent PID: %d\n", getppid());

        printf("Executing: ls -l\n");

        execlp("ls", "ls", "-l", NULL);

        /* This runs only if exec fails */
        perror("exec failed");
        exit(EXIT_FAILURE);
    }
```

```
    }
    else {
        /* Parent process */
        printf("\nParent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);
        printf("Parent waiting for child...\n");

        waitpid(pid, &status, 0);

        if (WIFEXITED(status)) {
            printf("\nChild exited with status: %d\n",
                   WEXITSTATUS(status));
        }
        else {
            printf("\nChild terminated abnormally.\n");
        }
```

```
        printf("Parent process completed.\n");
    }

    return 0;
}
```

OUTPUT:

```
root@DUDDUKURIs:~/skill/week-6# nano main2.c
root@DUDDUKURIs:~/skill/week-6# gcc main2.c -o main2
root@DUDDUKURIs:~/skill/week-6# ./main2
Session 6 - Process Execution

Parent Process
Parent PID: 551
Child PID: 552

Parent waiting for child...
Child Process
Child PID: 552
Parent PID: 551
Executing: ls -l
total 40
-rwxr-xr-x 1 root root 16184 Aug 15 04:34 main
-rw-r--r-- 1 root root  1337 Aug 15 04:34 main.c
-rwxr-xr-x 1 root root 16352 Aug 15 04:39 main2
-rw-r--r-- 1 root root  1198 Aug 15 04:38 main2.c

Child exited with status: 0
Parent process completed.
root@DUDDUKURIs:~/skill/week-6#
```